



An open-source Python library for asset management

Alexis Fremond

William Grison

What is ReLife ?

ReLife is an open-source library that leverages stochastic modeling to guide decision-making in asset management.

Problems solved by ReLife :

- When to maintain, replace or repair an asset ?
- Based upon which criterions ?
- What budget ?
- How to manage the stocks ?



Related work :

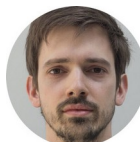
ReLife implements risk analysis as stated in IEC 63223-2, a risk-informed decision-making process



Core team



Thomas Guillon
Math expert, creator and first developer



Alexis Fremond
Math expert and developer



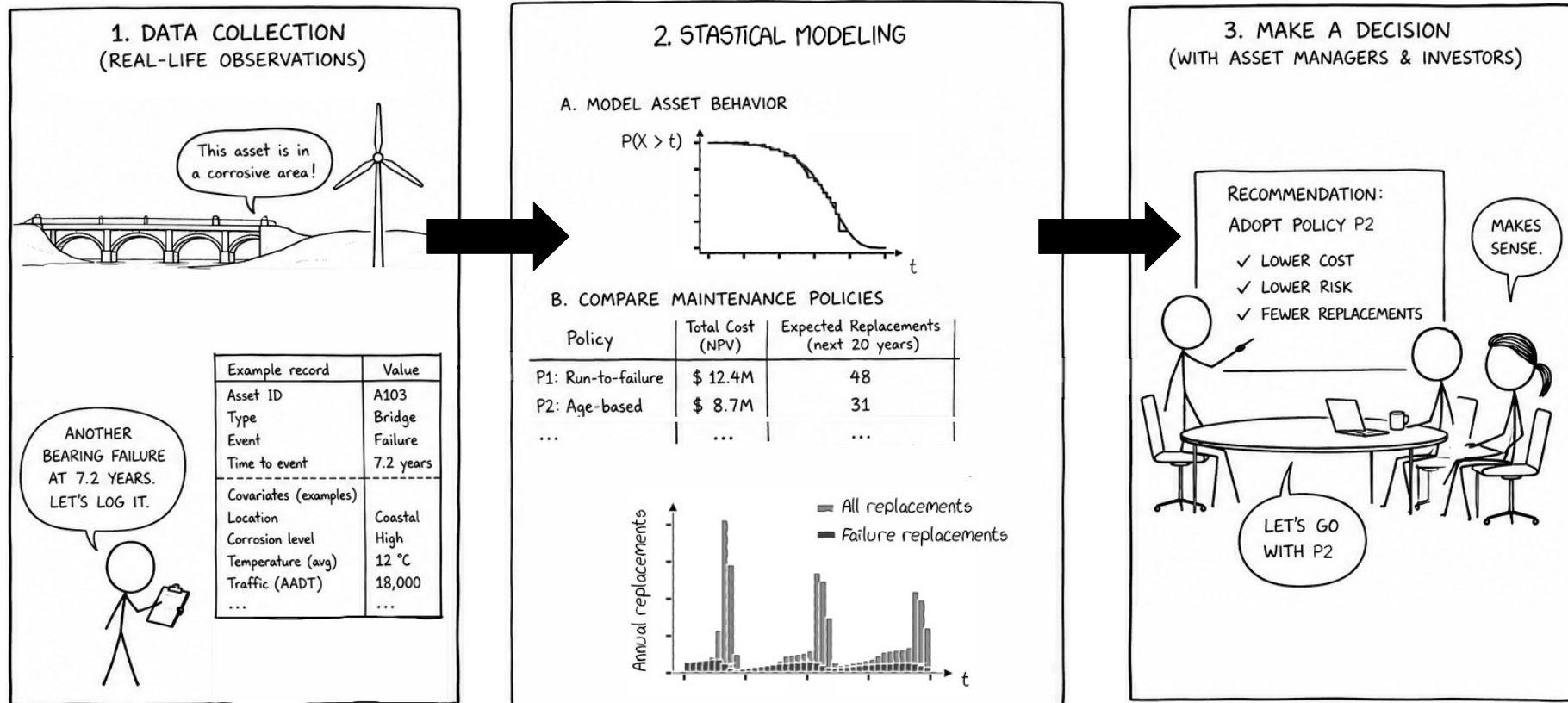
William GRISON
Main developer and engineer

Language :  python™

Built upon :  NumPy 

License :  THE ASF Apache 2.0

The mindset



Demo: age based replacements for circuit breakers

Functionalities overview



<http://opensource.rte-france.com/relife/>

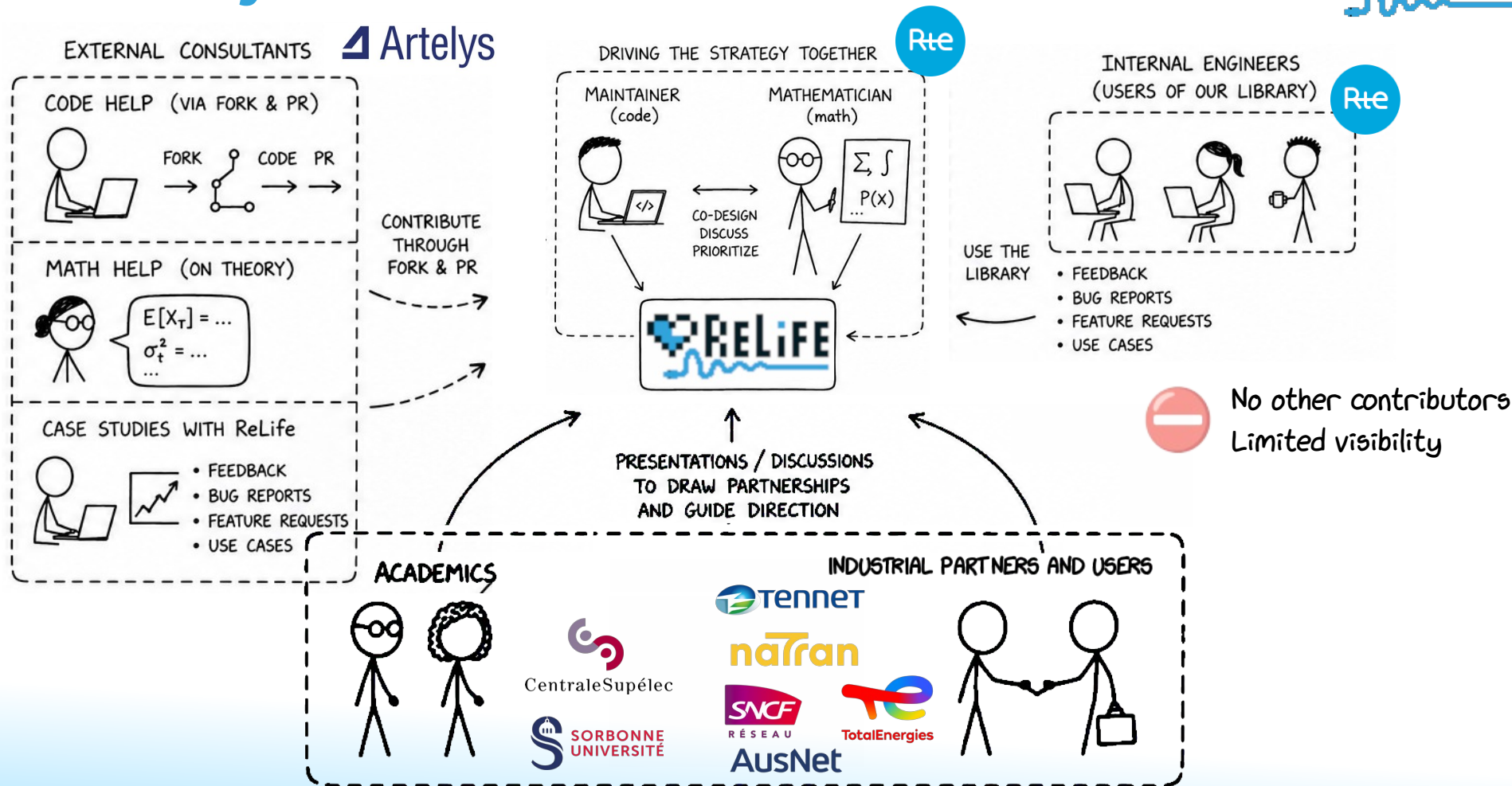
- Non parametric estimators.
- Parametric lifetime models with or without covariates.
- Semi-parametric regression (Cox regression).
- Counting process for recurrent events (non-homogeneous Poisson process).
- Renewal processes, with or without rewards.
- Renewal equation solver.
- N-dimensionnal Lebesgue-Stieltjes integration

About simulation

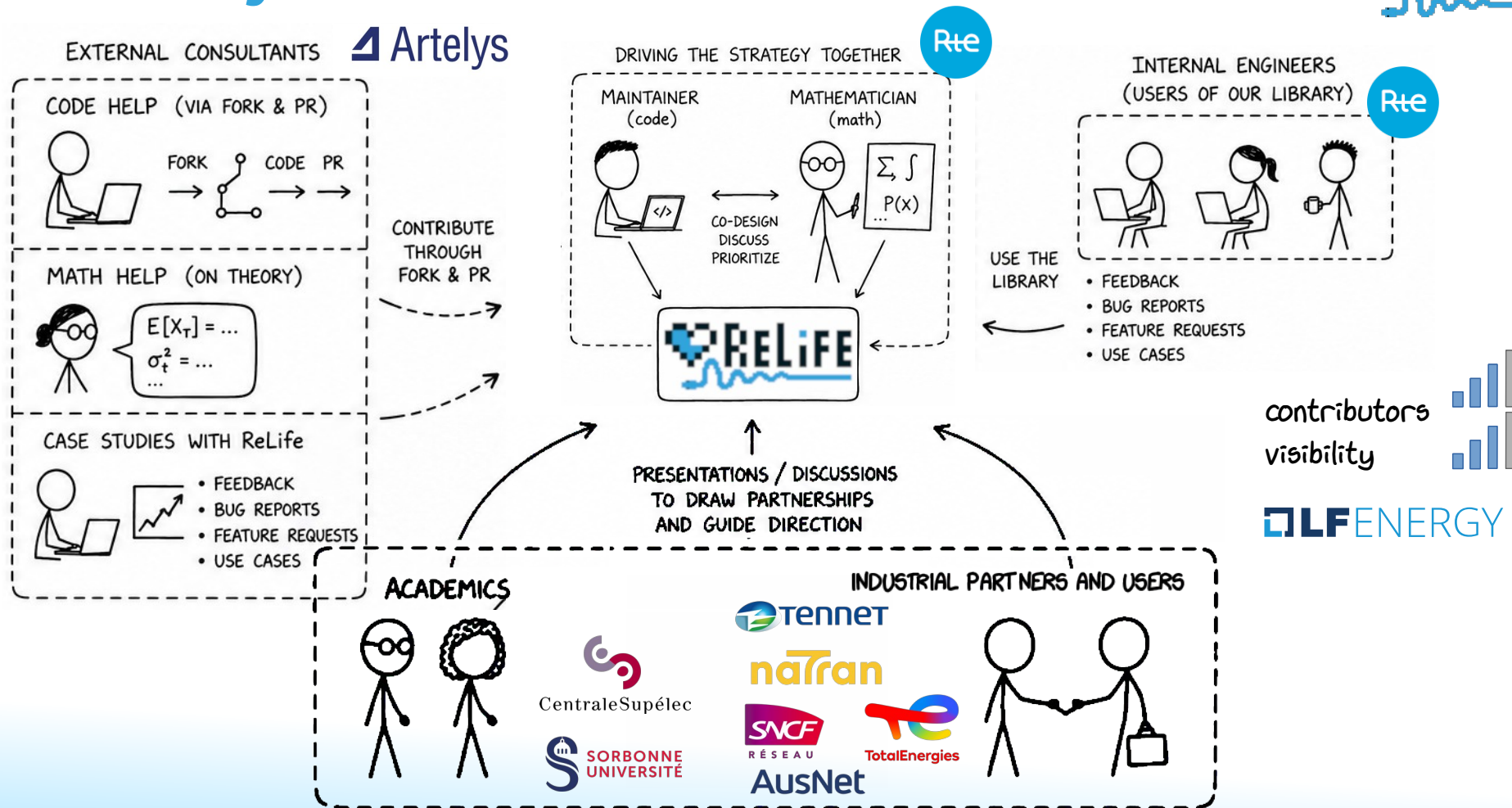
ReLife is **not a simulator ! It is a statistical toolbox.**

But it can simulate... To study the influence of data changes, to test fonctionnalities, maybe more ?

Our ecosystem



Our ecosystem



Upcoming releases



Short-term

End of 2026

- `stochastic_processes.GammaProcess` **e.g. deterioration process**
- **enhanced version of** `stochastic_processes.NonHomogeneousPoissonProcess` (fitting process stability)
- a better documentation (doctests, FAQ, end-to-end user guides).

Long-term

By 2027

- Orange Data Mining extensions - a GUI
- Virtual age processes, **e.g. imperfect repairs**
- Mixture models
- Competing risks models



Research projects

- Causal inference
- **Maybe others ?**



Thank you :)

Questions ?